

Module 12: Unit 2 AKTU PYQs and Solved Query Bank

AKTU Unit 2 Past 5-Year Solved Examination Papers

Relational Algebra vs SQL Dual Query Bank (Suppliers-Parts, Library, Student-Course)

1. Relational Algebra PYQ Solutions

Suppliers(sid, sname, city) | Parts(pid, pname, color)
Catalog(sid, pid, cost)

Q1: Suppliers of RED parts (AKTU 2021-22)

$\pi[sname](Suppliers \bowtie \pi[sid](Catalog \bowtie \sigma[color='RED'](Parts)))$
Joins Catalog with filtered red parts, then projects supplier names.

Q2: Suppliers who supply ALL parts (AKTU 2022-23)

$\pi[sname](Suppliers \bowtie (\pi[sid, pid](Catalog) \div \pi[pid](Parts)))$
Applies Division Operator ($\div$) to solve universal quantification.

Q3: Suppliers of at least 1 Red AND 1 Green part

$\pi[sname](Suppliers \bowtie (\pi[sid](Cat \bowtie \sigma[color='RED'](P)) \cap \pi[sid](Cat \bowtie \sigma[color='GREEN'](P))))$
Uses Set Intersection ($\cap$) of two supplier ID projections.

Q4: Suppliers who supply NO Red parts (AKTU 2023-24)

$\pi[sname](Suppliers) - \pi[sname](Suppliers \bowtie \pi[sid](Catalog \bowtie \sigma[color='RED'](Parts)))$
Uses Set Difference ($-$) to subtract red part suppliers from all suppliers.

2. SQL Query Dual Solutions

SQL Q1: Suppliers of RED parts (Join Approach)

```
SELECT DISTINCT S.sname FROM Suppliers S
JOIN Catalog C ON S.sid=C.sid JOIN Parts P ON C.pid=P.pid
WHERE P.color = 'RED';
```

SQL Q2: Suppliers of ALL parts (NOT EXISTS / Group By)

```
SELECT S.sname FROM Suppliers S JOIN Catalog C ON S.sid=C.sid
GROUP BY S.sid, S.sname
HAVING COUNT(DISTINCT C.pid) = (SELECT COUNT(*) FROM Parts);
```

Elegant COUNT aggregation approach solves relational division!

SQL Q3: Library Database (Book Issued by Student)

```
SELECT S.Name, B.Title FROM Student S
JOIN Issue I ON S.RollNo=I.RollNo JOIN Book B ON I.ISBN=B.ISBN;
```

SQL Q4: Students who borrowed NO books

```
SELECT Name FROM Student WHERE RollNo NOT IN (SELECT RollNo FROM Issue);
```

Uses NOT IN anti-join subquery to isolate zero-issue students.

Exam Proven Mastery: Relational Algebra numericals and SQL dual query solutions from the last 5 years of AKTU examinations.

1. Suppliers-Parts Relational Algebra Solutions

```
-- Q1: Suppliers of RED parts:
 $\pi[sname](Suppliers \bowtie \pi[sid](Catalog \bowtie \sigma[color='RED'](Parts)))$ 

-- Q2: Suppliers of ALL parts (Division):
 $\pi[sid, pid](Catalog) \div \pi[pid](Parts)$ 

-- Q3: Suppliers of NO Red parts (Set Difference):
 $\pi[sid](Suppliers) - \pi[sid](Catalog \bowtie \sigma[color='RED'](Parts))$ 
```